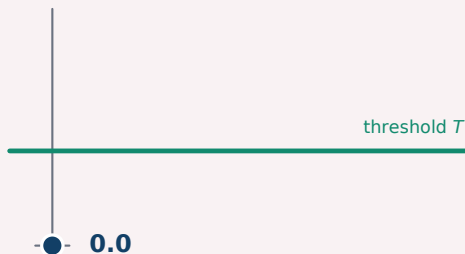


# A non-detection is an upper bound, not a zero

The same measurement, the same legal threshold, read two ways

## What the record stores

a number, indistinguishable from a measured zero

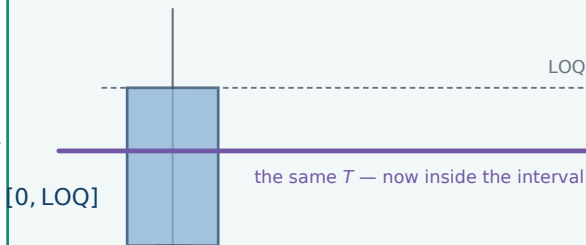


**Compliant**

one of two possible verdicts

## What the measurement established

an interval: the analyte may be present below the limit



**Cannot be determined**

a verdict the two-valued model cannot express

CENSO binds the quantification limit to the analytical run rather than to the instrument, so the interval survives into the data.

**45%**

of 33 million European samples  
lie below the quantification limit

**44%**

of assessments against a European  
standard cannot be determined  
from the record as reported

**40%**

of assessments against a European  
standard use a method failing the  
legal LOQ criterion (2009/90/EC)

EEA Waterbase: 4.2 million river station-years, 637 substances, 37 countries.

The reporting defect was repaired — station-years declaring neither flag nor limit reach 0 % in 2013 and stay there. The analytical one was not: 46 % in 2006, 48 % in 2024.